Policy-driven HR case orchestration

Design overview for the ServiceNow take-home · Raghav Munjal · November 2026 · Draft for discussion

Summary

HR teams receive a steady flow of questions about leave, benefits and payroll. Most of them are answered by a policy document that already exists, but finding the right policy, routing the case to the right team and writing a reply still takes people time.

This prototype handles HR cases with rules that HR can change without code, and uses an AI agent to help at each step. The agent reads the case, classifies it, finds the relevant policy text, and drafts a reply that quotes the policy word for word. ServiceNow stays in charge: it applies the routing rules, writes every change, keeps the audit trail, and asks a person to approve anything critical, such as closing a case.

Goals

- Route cases with configurable policy: an HR-owned rules table, so HR can change routing by editing a row.
- Use AI to classify cases, search policies and draft answers, with every factual claim backed by an exact quote.
- Let HR staff ask the AI for help inside a case, pointing it at a specific document.
- Keep a person in control of critical actions, and record who approved what.
- Design for a realistic policy library: hundreds of long documents, not a handful of short ones.

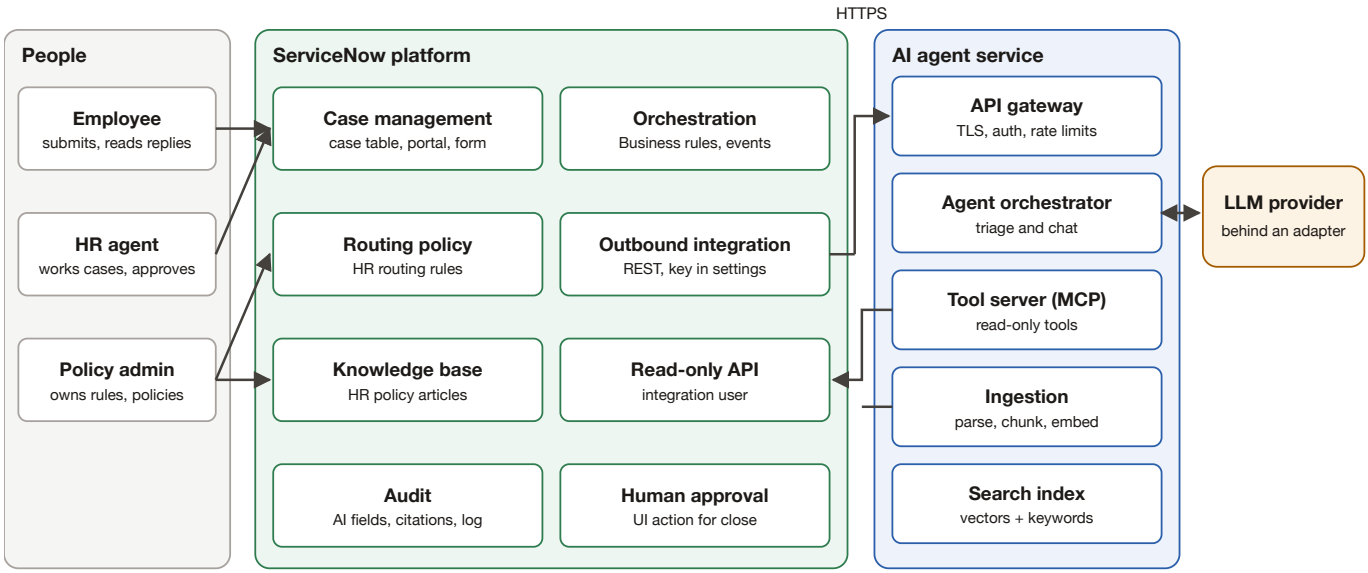
Not in scope for the prototype

Sensitive case handling (for example harassment or medical cases), personal data redaction, production hosting, multi-language support, and a dedicated chat panel. Each is covered in "Future work" so the path forward is clear.

How people use it

1. **An employee submits a case** on the portal with a title and description.
2. **The AI classifies it**: department, topics, priority and severity. Routing rules then pick the assignment group and assignee.
3. **The AI searches HR policies** using retrieval over the knowledge base.
4. **The employee gets an AI suggestion** on the case that quotes the exact policy text, as long as the AI is confident and can cite every claim. Otherwise no suggestion is posted and the case goes to people as usual.
5. **HR staff can tag `@ai`** in a comment or work note, point it at a document, and get a cited answer.
6. **A person approves critical actions.** The AI can propose closing a case, but only an HR agent can approve it.

Architecture



The design has three parts.

ServiceNow is the system of record. It holds the cases, the routing rules, the HR policy knowledge base and the audit trail. Business rules and events call the AI agent service, apply its recommendation through the routing rules, and post replies. A UI action gives HR agents the approval step for closing a case.

The AI agent service is a small, containerized service outside ServiceNow. It runs the agent loop, exposes read-only tools over the Model Context Protocol (MCP), and keeps a search index of the policy library. It reads from ServiceNow with a read-only integration user and never writes to a case. It returns a recommendation, and ServiceNow decides what to do with it.

The LLM provider is accessed through an adapter, so the model can be swapped without changing the rest of the system. A hosted model with tool calling and citation support is the default. Moving to Now Assist on a licensed instance would replace this adapter, not the design.

Why an external service

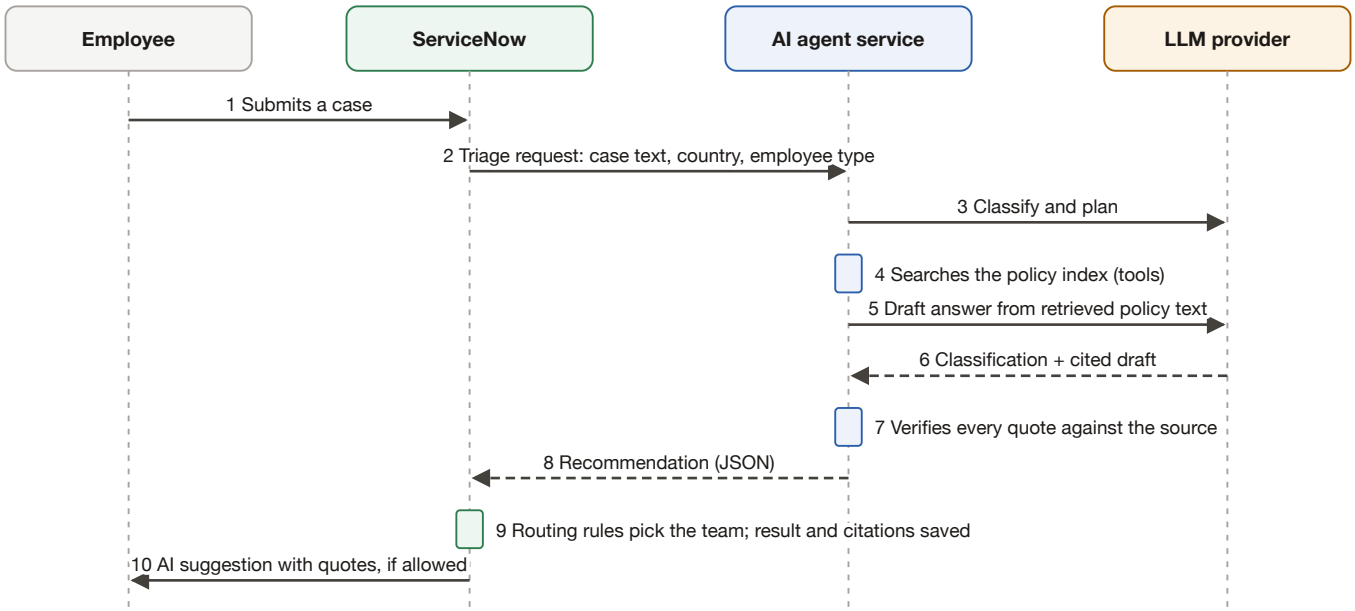
Personal developer instances don't include Now Assist or AI Agent Studio, and good retrieval over hundreds of long documents needs a proper search index. Keeping that work in a separate service also keeps the ServiceNow app small and upgrade-safe. The integration uses standard patterns (outbound REST with the key kept out of source, and the Table API with a least-privilege user), so it could be pointed at a different AI backend later.

Key design decisions

Decision	Why	Trade-off
The AI recommends; ServiceNow decides and writes	Keeps control, audit and permissions in the platform. The AI service needs no write access	Every action needs logic in ServiceNow to apply the recommendation

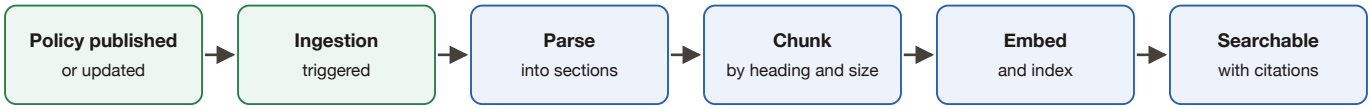
Routing lives in an HR-owned rules table	HR changes routing in the UI without code, and deployments never overwrite their edits	Not Decision Builder, which the build tooling cannot define as code; switching later changes one script
Every factual claim must quote the policy exactly	Employees and HR can check the answer, and hallucinated policy never reaches an employee	Some questions get no suggestion when the policy can't be quoted
The knowledge base is the source of truth; the search index is a copy	Policy owners keep using ServiceNow Knowledge; the index can be rebuilt any time	Ingestion has to keep the index in sync
Hybrid search with reranking	Keywords catch exact terms and form names, while vector search catches paraphrases	More components than keyword search alone
Human approval for closing	Closing affects the employee directly and is hard to undo	One more click for HR
Fail safe to people	Any AI failure sends the case to manual triage instead of blocking it	Cases can reach people without AI help during an outage
Provider-neutral LLM access	Avoids lock-in and keeps a clear path to Now Assist	Provider-specific features sit behind the adapter

Flow of a new case



If the service is unreachable, times out or returns something invalid, the case goes to a Manual Triage group and the failure is noted on the case. If the AI isn't confident, or any claim lacks a citation, routing still happens but no suggestion is posted.

Knowledge and retrieval



When a policy article is published, updated or retired in ServiceNow, the AI service re-ingests it. A nightly check compares versions to catch anything missed. Retired articles are kept for audit but never searched.

Chunking

Articles are split along their headings first, so a chunk never mixes two sections, and then into pieces of a few hundred words with a small overlap, cut only at sentence boundaries. Tables are kept whole where possible, and the section path ("Parental leave > Germany > Eligibility") travels with each chunk so it keeps its context. Each chunk remembers exactly where it came from in which version of the article.

Search

Searches always filter to active policies for the employee's country (or global ones). The service runs a vector search and a keyword search, merges the results, reranks them and keeps the best few. If nothing is relevant enough, the agent says it couldn't find a policy instead of guessing.

Citations

The agent drafts its answer only from the retrieved text. Every factual sentence must carry a quote, and the service checks each quote word for word against the stored source before anything is returned. ServiceNow saves each citation with the article version, so you can always see what the AI relied on, even after the policy changes. An AI suggestion looks like this:

> You can take up to three years of parental leave per child, and you need to request it in writing at least seven weeks before it starts. [1][2]
>
> [1] Parental leave (Germany), Eligibility: "Employees in Germany are entitled to up to 3 years of parental leave per child."
> [2] Parental leave (Germany), Eligibility: "Leave must be requested in writing at least 7 weeks before it starts."

What the AI can and can't do

The AI agent can	The AI agent can't
Classify a case and explain its reasoning	Write to a case, assign it or close it
Search policies and read a specific section	Choose the assignment group (the routing rules do)
Draft a reply in which every claim is quoted	State a policy it can't quote
Answer `@ai` questions from HR staff	Change routing rules or policy articles
Propose closing a case, with a resolution	Act on instructions written inside a case

Technology choices

Area	Choice	Why
ServiceNow app	Scoped application built with the ServiceNow SDK, kept in Git	Upgrade-safe and under source control
Orchestration and policy	Business rules, events and script actions; HR-owned routing rules table	Runs off the user's transaction; routing is data HR edits
Platform logic	Script Includes, Business Rules, `RESTMessageV2`; service URL and key in admin-only system properties	Standard, reusable integration pattern
User interface	Service Portal record producer, case form, UI action for approval	Out-of-the-box experiences
Service language	Python	Strongest ecosystem for agents, retrieval and evaluation
API server	FastAPI with Pydantic models	Typed request and response contracts, async, automatic API docs
Agent loop	LangGraph	An explicit graph (classify, retrieve, draft, verify) with bounded tool calls, easy to test and reason about
LLM access	Provider adapter over the provider's official SDK	Model can be swapped by configuration
Tools	Model Context Protocol, official Python SDK	Standard tool interface the agent and other clients can share
Search index	PostgreSQL with pgvector and built-in full-text search	Vectors and keywords in one database
Embeddings and reranking	Open-source ONNX models run inside the service (FastEmbed): an embedding model and a cross-encoder reranker	No extra vendor, policy text stays in our infrastructure
Parsing and chunking	BeautifulSoup and a heading-aware splitter	Keeps policy structure intact
Gateway	Caddy reverse proxy	Automatic TLS, routing and rate limits
Packaging	Docker Compose	One command to run the whole service
Testing	pytest, a retrieval evaluation set, ServiceNow ATF where time allows	Covers code, retrieval quality and platform behaviour

Security and governance

- **Least privilege.** The AI service reads cases and HR articles through a read-only integration user. It can't write anything in ServiceNow.

- **Authenticated calls.** ServiceNow calls the service over HTTPS with an API key kept in an admin-only system property, optionally with signed, time-stamped requests to prevent replays.
- **Small surface.** Only the gateway is exposed, and only for the endpoints ServiceNow needs. The index, models and tools stay internal.
- **Prompt injection.** Case text is treated as data, not instructions. Even a successful injection can't change a case, because the agent has no write access and closing needs approval.
- **Audit.** Every AI result, citation and approval is stored on the case in ServiceNow.
- **Data.** The prototype uses synthetic employees, cases and policies only.

Testing

Scenario	Expected result
Routine benefits question	Routed to Benefits, with a suggestion that quotes the benefits policy exactly
Same leave question from two countries	Each answer cites its own country's policy
Vague request	Low confidence: no suggestion, routed to Manual Triage
HR edits a routing rule	The next case routes differently, with no code change
A policy article is updated	New answers quote the new text; older citations keep the old version
HR tags `@ai` and points at an article	The reply cites that article
The AI proposes closing a case	The case stays open until an HR agent approves
AI service unavailable	Manual Triage, with the failure noted on the case
Case text tries to override the AI's instructions	No effect

Retrieval quality is measured separately with a set of 30 to 50 questions, each with a known correct policy section. The target is that the right section appears in the top five results and that every quote matches its source.

Delivery plan

1. **Foundations.** Confirm instance capabilities and connectivity between ServiceNow and the AI service.
2. **ServiceNow app.** Case model, portal form, roles, groups and the routing rules.
3. **Policy library.** A synthetic set of 50 to 100 long policies across several countries.
4. **Retrieval.** Ingestion, chunking, the search index and the retrieval evaluation.
5. **Triage end to end.** Classification, routing and cited suggestions.
6. **`@ai` assistance and close approval.**
7. **Hardening.** Failure paths and injection tests.
8. **Hand-off.** This document finalized, a recorded walkthrough and test evidence.

Future work

- **Sensitive case handling.** A configurable sensitivity check before any AI call, with a restricted queue and access controls. This is the first step before any real data.
- **Personal data redaction** before case text leaves the platform.
- **Now Assist and AI Agent Studio.** On a licensed instance, replace the provider adapter with Now Assist skills and agents, and consider ServiceNow AI Search for retrieval.
- **Richer policy governance:** versioned routing policies with owners and effective dates, and per-category limits on what the AI may do automatically.
- **Production operations:** managed hosting, tracing dashboards, a full ATF suite and CI/CD.

Assumptions

- The prototype is built on a personal developer instance. That instance doesn't include HR Service Delivery, Now Assist or AI Agent Studio.
- Because HR Service Delivery isn't available, cases live in a custom table modeled on the HR case fields. In production this would be the standard HR case table.
- Calling an external LLM provider is acceptable, because all data is synthetic.
- A small service outside ServiceNow is acceptable for the AI and retrieval work, reachable from the instance over HTTPS.
- The prototype's hosting is for demonstration only and isn't meant to reflect production hosting.
- A synthetic library of 50 to 100 long policies across a few countries is a fair stand-in for a real policy library.
- English only.
- Closing a case is the only action that needs human approval in the prototype. The same pattern extends to others.
- Sensitive cases are out of scope for the prototype and handled by the future-work design above.